
Technical and Business Writing

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Importance of Reading Scientific Papers

- **Foundation of academic knowledge:** the primary way researchers communicate discoveries.
- **Staying updated** – New findings shape advancements in your field.
- **Conducting literature reviews** – Understanding past studies is crucial for building upon existing work
- **Developing critical thinking** – Analyzing methods, results, and conclusions
- **Applying research effectively** – Proper interpretation of data ensures accurate application in real-world scenarios.

The IMRD Structure

Most scientific journals use a conventional structure that helps readers navigate complex information efficiently.



Introduction

Provides context and rationale



Methods

Describes experimental design



Results

Presents findings and data



Discussion

Interprets results and implications

Understanding Each Section

Abstract

Offers a quick overview of the study's objectives, methods, key findings, and conclusions.

Reading the abstract first helps determine whether the paper is relevant, but since abstracts lack full details, they should never be relied upon exclusively.

Introduction

Creates interest and provides understanding by leading from broad information (what is known) to specific (what is unknown) to focal point (what question was asked and answered).

Understanding Each Section

Methods

- A **blueprint** for how the research was conducted.
- Tells what experiments were done to answer questions.
- Often difficult due to technical language.
- Read carefully to understand experimental design and evaluate validity

Result & Discussion

- Results (what happened) contain findings and references to visuals.
- Discussion (why it matters) places work in context, provides clear answers to questions, and explains how results support conclusions.

Generate Questions While Reading

Before & During Reading

- Who are these authors? What journal is this?
- Have I understood all key terminologies?
- Am I spending time on important parts?
- Is there someone to discuss confusing sections?

Generate Questions While Reading

After Reading

- What problem does this research address?
- What are the specific findings?
- Are findings supported by evidence?
- How do results relate to other work?

Take Notes as You Read

Develop a template

Record key elements

Save time later

How to Take Notes

Complete Citation

Author(s), date, title, journal,
volume, issue, pages

Keywords

General subject and specific
subject

Hypothesis

What question did the
authors ask?

How to Take Notes

Methodology

What experiments were conducted?

Results

What were the key findings?

Summary

Key points in your own words

How to Take Notes

Context

How it relates to other work
in the field

Significance

Importance to the field and
your work

Important Figures

Brief descriptions and page
numbers

Three Essential Tasks

Comprehension: Understanding what the paper says and establishing the foundation

Evaluation: Critically assessing the validity and significance of claims

Synthesis: Generating new research ideas from existing work

Three Essential Tasks

Comprehension: Understanding what the paper says and establishing the foundation

Research Problem

- ✓ What crisis does the paper address?
- ✓ Is an existing paradigm challenged?
- ✓ What is the niche?

Contribution

What is new?

- ✓ New methodology,
- ✓ Algorithm
- ✓ Tool
- ✓ Proof technique, or evidence?

Three Essential Tasks

Evaluation: Critically assessing the validity and significance of claims

Is the research problem significant?

- ✓ Are problems artificial or practical?
- ✓ Does it build upon existing paradigms?

Are the contributions significant?

- ✓ Are authors aware of existing literature?
- ✓ Is it addressing a well-known open problem?

Are the claims valid?

- ✓ Has proper methodology been followed?

Three Essential Tasks

Synthesis: Generating new research ideas from existing work



What is the crux?

Identify the fundamental challenge at the heart of the research problem



Alternative approaches?

Can the problem be addressed differently or substantiated another way?



Counter-arguments?

What's a good argument against the authors' case?



Strengthen results?

Can findings be extended, improved, or applied to another context?

Three Essential Tasks

Synthesis: Generating new research ideas from existing work



Open problems?

What unresolved questions does this work raise?



Do differently?

If you conducted this research, how would your approach differ?